MODULE 3: PROGRAMMING FOR BIOLOGICAL DATA

Sanger Sequencing Analysis

Chromatograms, Quality Scores & sangeranalyseR

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Learning Objectives

Sanger

Understand the principles of Sanger sequencing and chain termination.

Chromatograms

Interpret trace files (.ab1) and identify common artifacts.

Phred

Calculate and apply Phred quality scores (Q20, Q30).

sangeranalyseR

Process, trim, and assemble sequences using Bioconductor.

Part 1: The Anatomy of a Chromatogram

From Fluorescent Peaks to DNA Sequence

Sanger Sequencing History

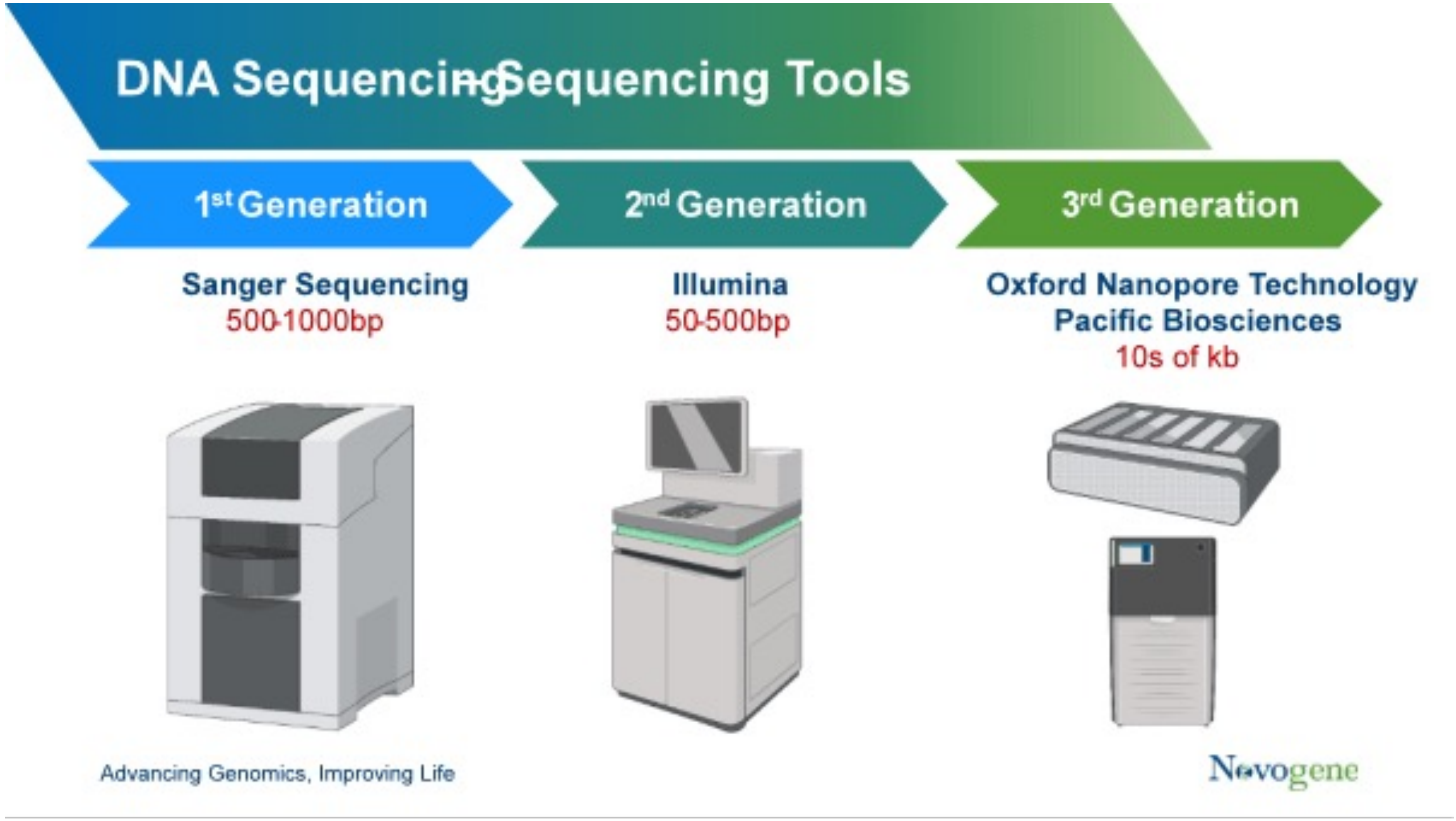
The Gold Standard

Developed by **Frederick Sanger** in 1977.

- Won Nobel Prize in Chemistry (1980).
- Known as "Chain Termination" method.

Modern Use

- Confirming mutations found by NGS.
- Sequencing single genes (e.g., 16S rRNA).
- Validating cloning results.



How It Works

The Ingredients

- **Template DNA**
- **Primer**
- **DNA Polymerase**
- **dNTPs** (Normal nucleotides)
- **ddNTPs** (Fluorescent Terminators)

The Process

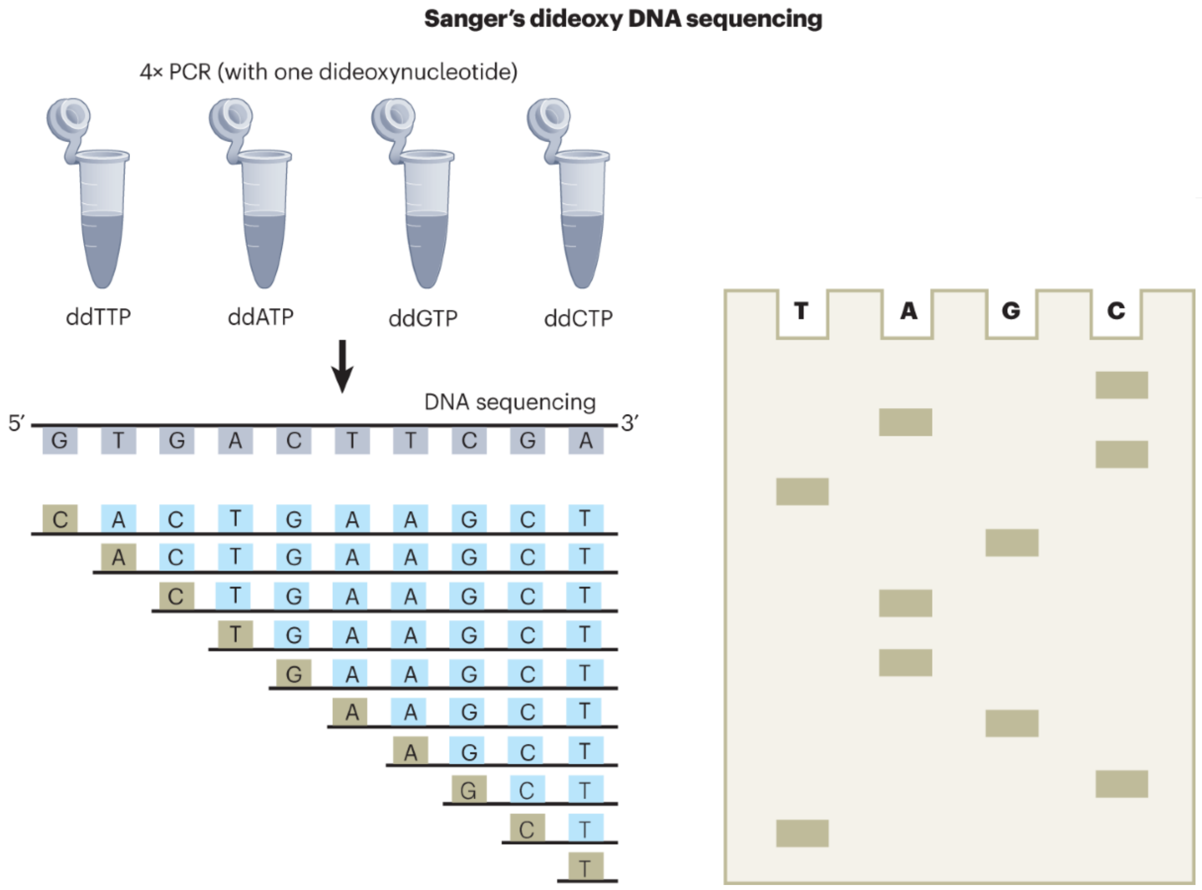
1. Synthesis starts.
2. Random incorporation of a **ddNTP** stops the chain.
3. Fragments of all lengths are produced.
4. Separated by **Capillary Electrophoresis**.
5. Laser detects the color of the last base.

Dideoxynucleotides (ddNTPs)

The key to the method is the missing 3'-OH group.

Without it, the DNA polymerase cannot add the next nucleotide.

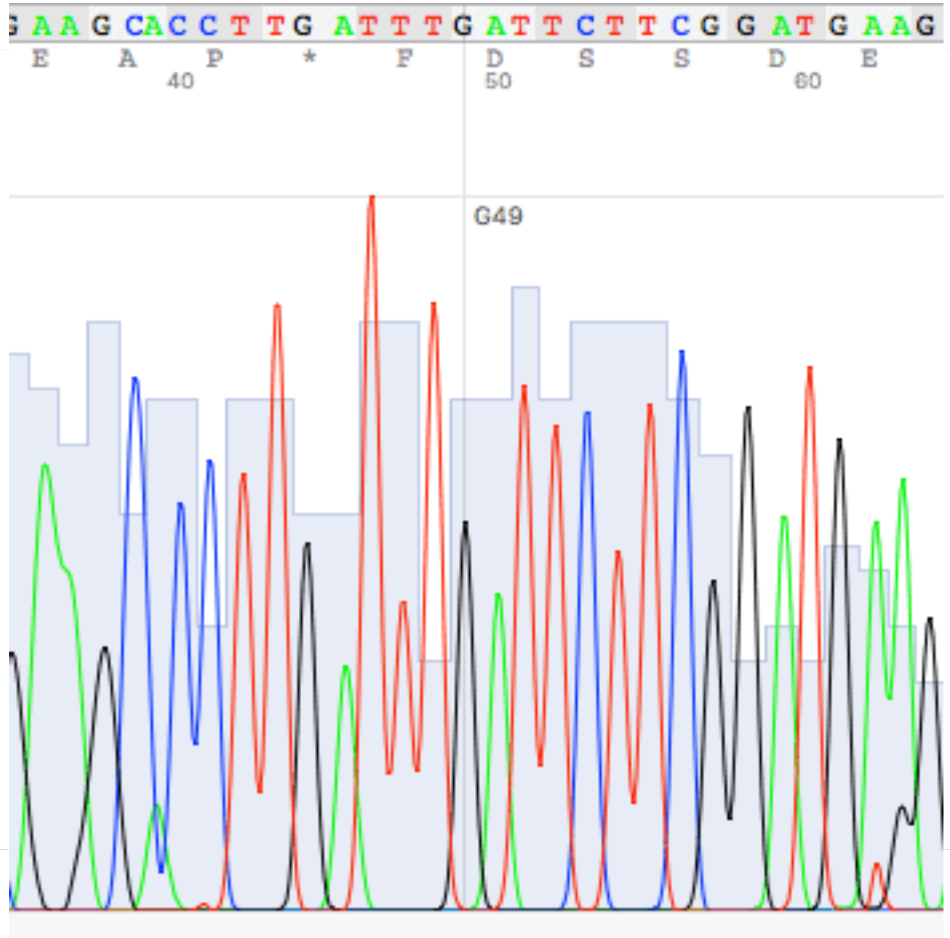
Terminator	Color
ddATP	Green
ddTTP	Red
ddGTP	Black
ddCTP	Blue



What is a Chromatogram?

The raw data file (.ab1) contains the fluorescence signal trace.

- **X-axis:** Time / Position.
- **Y-axis:** Fluorescence Intensity.
- **Peaks:** Represents a base call.



Reading a Chromatogram

✓Good Trace

- Sharp, defined peaks.
- Even spacing.
- Low background noise.
- High signal intensity.

✗Bad Trace

- Overlapping "mushy" peaks.
- Variable spacing.
- High background baseline.
- Mixed signals (two peaks at once).

Common Chromatogram Issues

Issue	Location	Cause
Dye Blobs	Start (5')	Unincorporated dye terminators not removed.
Signal Decay	End (3')	Depletion of reagents; long fragments difficult to resolve.
Mixed Peaks	Variable	Heterozygous mutation or sample contamination.

Phred Quality Scores

A standardized metric to quantify base call confidence.

$$Q = -10 \times \log_{10}(P_{error})$$

Developed by Phil Green (University of Washington).

Interpreting Phred Scores

Q Score	Error Probability	Accuracy
Q10	1 in 10	90%
Q20	1 in 100	99%
Q30	1 in 1,000	99.9%
Q40	1 in 10,000	99.99%

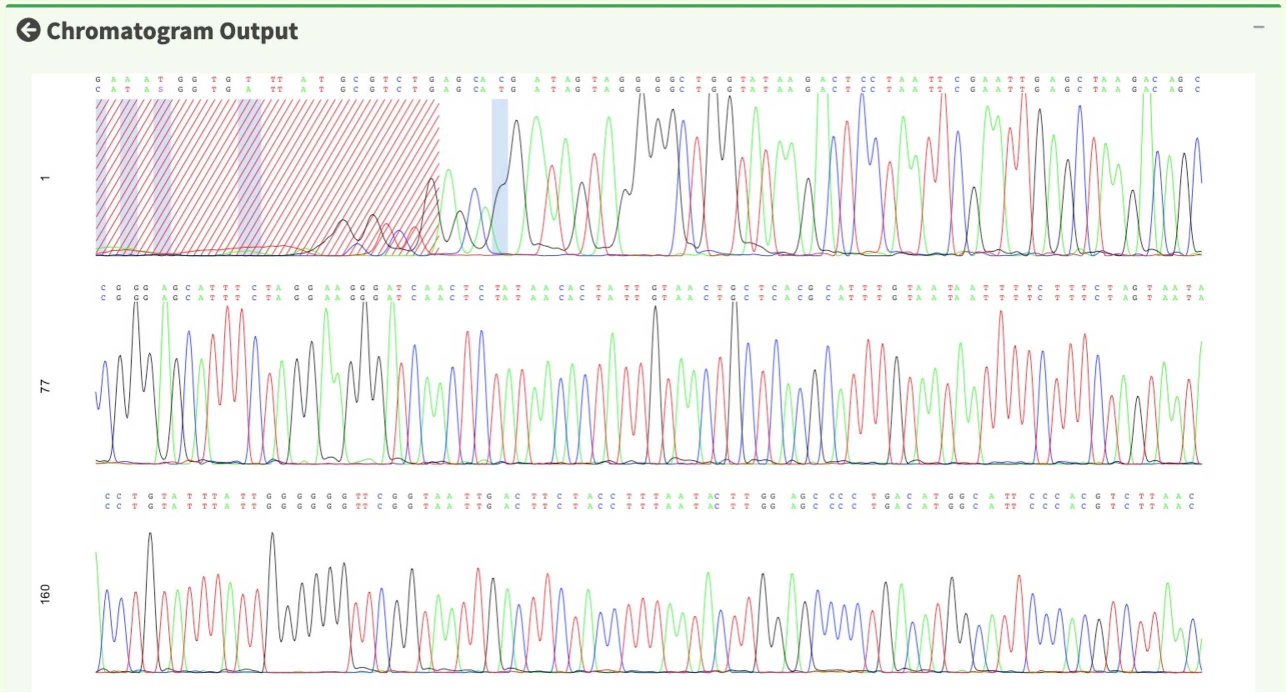
Rule of Thumb: Q20 is the minimum for reliable data.

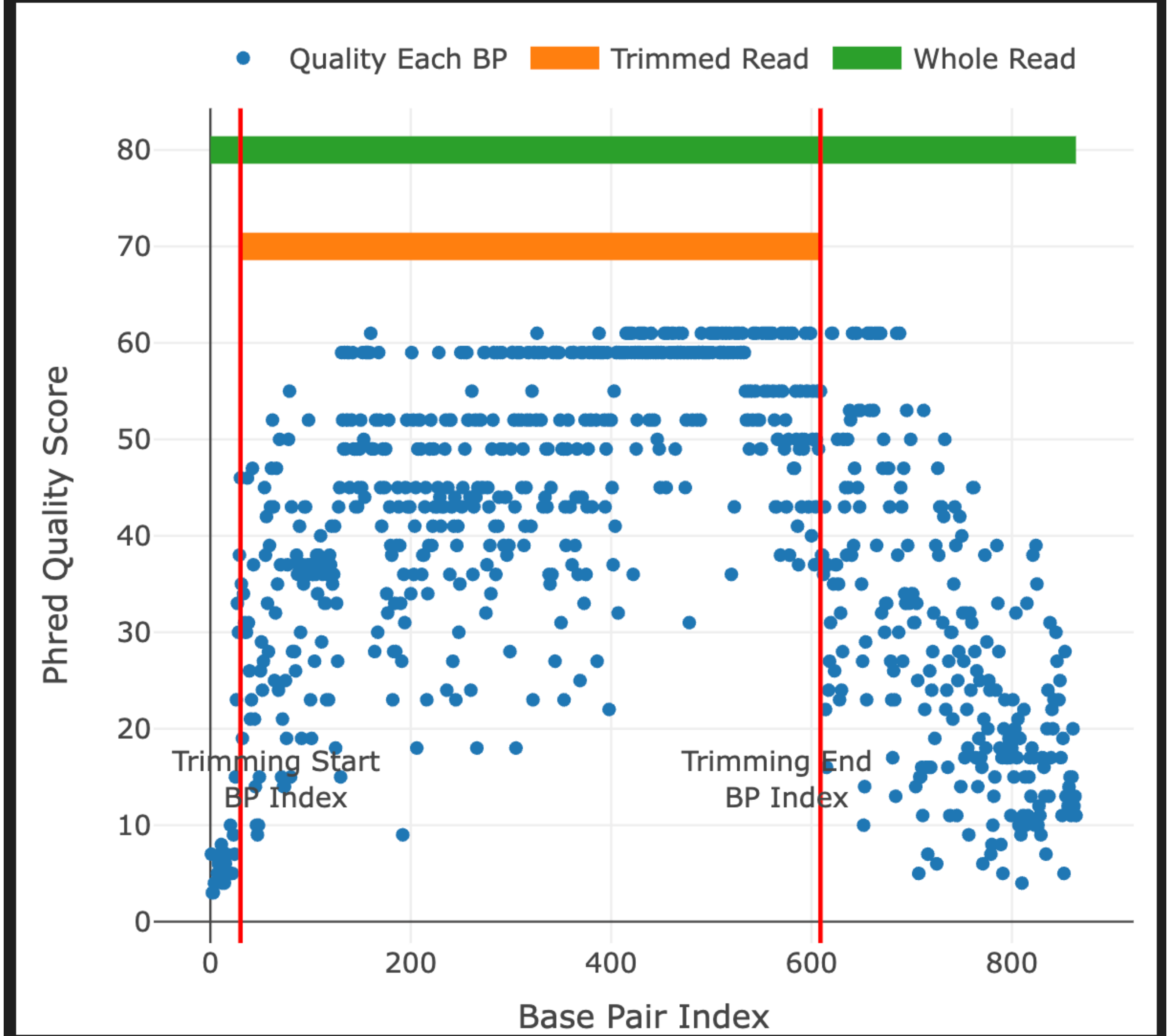
Typical Quality Profile

The "Frown" Shape

Sanger reads typically show:

- **Start (5'):** Lower quality due to primer effects and dye blobs.
- **Middle:** High quality, reliable sequence.
- **End (3'):** Declining quality as signal weakens and resolution drops.





Quality Each BP
Trimmed Read
Whole Read
Phred Quality Score
80
70
60
50
40
30
20
10
0
Trimming Start
BP Index
Trimming End
BP Index
0
200
400
600
800
Base Pair Index

Why Quality Trimming?

Errors

Low-quality bases lead to false mutation calls.

Alignment

Noisy ends prevent accurate alignment to reference genomes.

Solution

Trim the unreliable ends to keep only the high-quality middle.

Samples

from previous the lesson

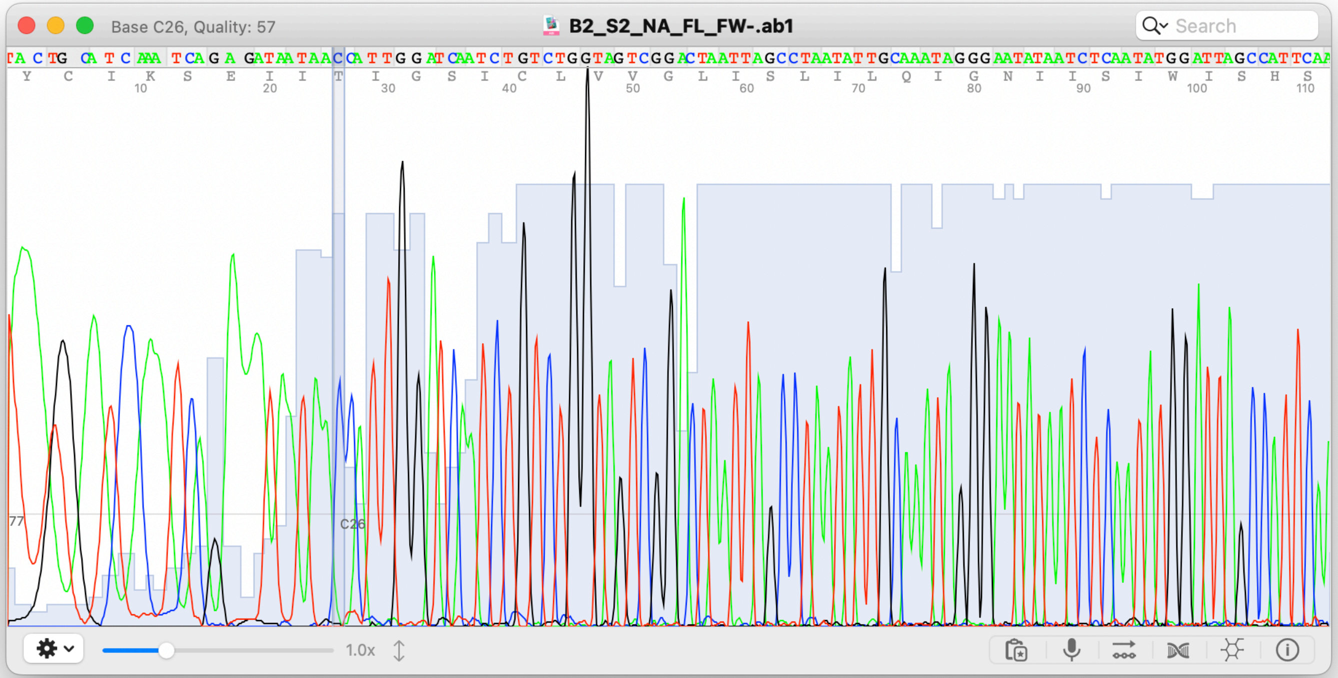
- We sent the samples of two group (Group B2, D2) for the reactions.
- Sanger sequencing of the HA and NA gene of one H1N1 and one H3N2 sample.
- HA (3 sequencing reactions): HA-universal Forward primer (HA-1F), H1 / H3 internal forward primer (336F), Universal reverse primer (NS-890R)
- NA (2 sequencing reactions): NA-universal forward and reverser primers (NA-1F, NA-1413R)

Name	Date Modified	Size	Kind
B2_S2_H1_HA_Int_FW-.ab1	19 Nov 2024 at 23:34	276 KB	ABI
B2_S2_HA_FL_FW-.ab1	19 Nov 2024 at 23:34	273 KB	ABI
B2_S2_HA_FL_RV-.ab1	19 Nov 2024 at 23:34	272 KB	ABI
B2_S2_NA_FL_FW-.ab1	19 Nov 2024 at 23:34	270 KB	ABI
B2_S2_NA_FL_RV-.ab1	19 Nov 2024 at 23:34	271 KB	ABI
B2_S3_H3_HA_Int_FW-.ab1	19 Nov 2024 at 23:34	274 KB	ABI
B2_S3_HA_FL_FW-.ab1	19 Nov 2024 at 23:34	249 KB	ABI
B2_S3_HA_FL_RV-.ab1	19 Nov 2024 at 23:34	267 KB	ABI
B2_S3_NA_FL_FW-.ab1	19 Nov 2024 at 23:34	273 KB	ABI
B2_S3_NA_FL_RV-.ab1	19 Nov 2024 at 23:34	274 KB	ABI
D2_S2_H1_HA_Int_FW-.ab1	19 Nov 2024 at 23:34	271 KB	ABI
D2_S2_HA_FL_FW-.ab1	19 Nov 2024 at 23:34	276 KB	ABI
D2_S2_HA_FL_RV-.ab1	19 Nov 2024 at 23:34	272 KB	ABI
D2_S2_NA_FL_FW-.ab1	19 Nov 2024 at 23:34	270 KB	ABI
D2_S2_NA_FL_RV-.ab1	19 Nov 2024 at 23:34	272 KB	ABI
D2_S3_H3_HA_Int_FW-.ab1	19 Nov 2024 at 23:34	277 KB	ABI
D2_S3_HA_FL_FW-.ab1	19 Nov 2024 at 23:34	271 KB	ABI
D2_S3_HA_FL_RV-.ab1	19 Nov 2024 at 23:34	273 KB	ABI
D2_S3_NA_FL_FW-.ab1	19 Nov 2024 at 23:34	277 KB	ABI
D2_S3_NA_FL_RV-.ab1	19 Nov 2024 at 23:34	275 KB	ABI

Sanger sequencing

AB1 files

- AB1 files are binary files produced by Applied Biosystems' Sanger sequencing machines.
- They store raw data from sequencing runs, including chromatograms and metadata.
- Contents of an AB1 File:
    - Chromatogram Data: Electropherogram peaks for A, T, C, G.
    - Base Calls: The called DNA sequence.
    - Quality Scores: PHRED scores for each base.
    - Metadata: Information about the sequencing run (e.g., instrument settings, sample info).



Sanger sequencing

AB1 files

- Reading AB1 files:
  - GUI-Based tools:
    - Tools with graphical interfaces are ideal for beginners or non-programmers.
  - Command line tools:
    - Ideal for advanced users and automation. Requires coding knowledge.

Sanger Data Analysis

	Free/Paid	Trace File Viewer	Multiple Sequence Alignment & Contig Assembly	Windows	Mac	Linux	In-browser
4Peaks	Free	✓	-	-	✓	-	-
Sequence Scanner	Free	✓	-	✓	-	-	-
Chromas Lite	Free	✓	-	✓	-	-	-
SnapGene Viewer	Free	✓	-	✓	✓	✓	-
GeneStudio Pro	Free	✓	✓	✓	-	-	-
UGENE	Free	✓	✓	✓	✓	✓	-
Chromaseq	Free	✓	✓	✓	✓	✓	-
BioLign (Note: See bottom half of page for description and download link)	Free	-	✓	✓	-	-	-
CAP3 (Note: Only for contig assembly of sequences in FASTA format)	Free	-	✓	-	-	-	✓
Geneious	Paid	✓	✓	✓	✓	✓	-
SeqMan Pro	Paid	✓	✓	✓	✓	-	-
Sequencher	Paid	✓	✓	✓	✓	-	-
Vector NTI Express	Paid	✓	✓	✓	✓	-	-
Chromas Pro	Paid	✓	✓	✓	-	-	-
CodonCode Aligner	Paid	✓	✓	✓	✓	-	-
DNA Baser	Paid	✓	✓	✓	-	-	-

Part 2: The sangeranalyseR Package

Automated Analysis in R

sangeranalyseR

Features

- **Bioconductor** package.
- Reads .ab1 files directly.
- Visualizes chromatograms and quality.
- Automated trimming (Mott's Algorithm).
- Contig assembly (Forward + Reverse).

sangeranalyseR's tutorial

Why sangeranalyseR

sangeranalseR is an R package that provides fast, flexible, and reproducible workflows for assembling your sanger seuqencing data into contigs.

It adds to a list of already widely-used tools, like Geneious, CodonCode Aligner and Phred-Phrap-Consed;. What makes it different from these tools is that it's free, it's open source, and it's in R.

Main features

- **Pure R environment**: As far as we know, this is the first package that allows end-to-end analysis of Sanger sequencing data in a pure R environment.
- **Automated data analysis**: Given appropriately-named input files, a lot of the data analysis can be automated. Once you've set up an appropriate workflow for your data, you can run it again in seconds.
- **Interactive Shiny apps**: Local Shiny apps mean you visualize the data at many levels, view chromatograms, and adjust things like trimming parameters.
- **Exporting and importing FASTA files**: sangeranalyseR is primarily designed with loading raw `ab1` files in mind, but it can also load sequencesin **FASTA** format. Aligned results and trimmed reads can be written into **FASTA** file format.
- **Thorough report**: A single command creates a comprehensive interactive HTML report that provides a huge amount of detail on the analysis.

https://sangeranalyser.readthedocs.io/en/latest/content/quickstart.html

Loading an AB1 File

library(sangeranalyseR) # Create SangerRead object my_read <- SangerRead(readFileName = "sample_1.ab1", readFeature = "Forward Read") # View summary my_read

Visualizing Quality

```
# Quality plot (Heatmap)
qualityBasePlot(my_read)

# View Chromatogram
chromatogram(my_read)
```

These functions generate interactive plots to inspect signal strength and base call confidence.

Extracting Sequences

Get raw sequence (Untrimmed) primarySeq(my_read, trim = FALSE) # Get trimmed sequence (Clean) primarySeq(my_read, trim = TRUE) # Get quality scores rawSeqQuality(my_read)

Mott's Trimming Algorithm

The Logic

1. Set an error probability limit (e.g., 0.05).
2. Calculate score: Limit - P_error.
3. Calculate cumulative sum across sequence.
4. Identify the substring with the maximum cumulative score.

The Result

This dynamic method trims poor quality regions from **both** ends while preserving the high-quality core.

Trimming Parameters

my_read <- SangerRead(readFileName = "sample.ab1", TrimmingMethod = "M1", # Mott's Algorithm M1TrimmingCutoff = 0.0001, # Stringency (Lower = Stricter) baseNumPerRow = 100 # Formatting)

Quality Statistics

Extract quality vector quals <- rawSeqQuality(my_read) # Calculate metrics mean(quals) # Average quality sum(quals >= 20) # Count of good bases length(quals) # Read length

Exporting to FASTA

Once trimmed, save your clean sequence for downstream analysis.

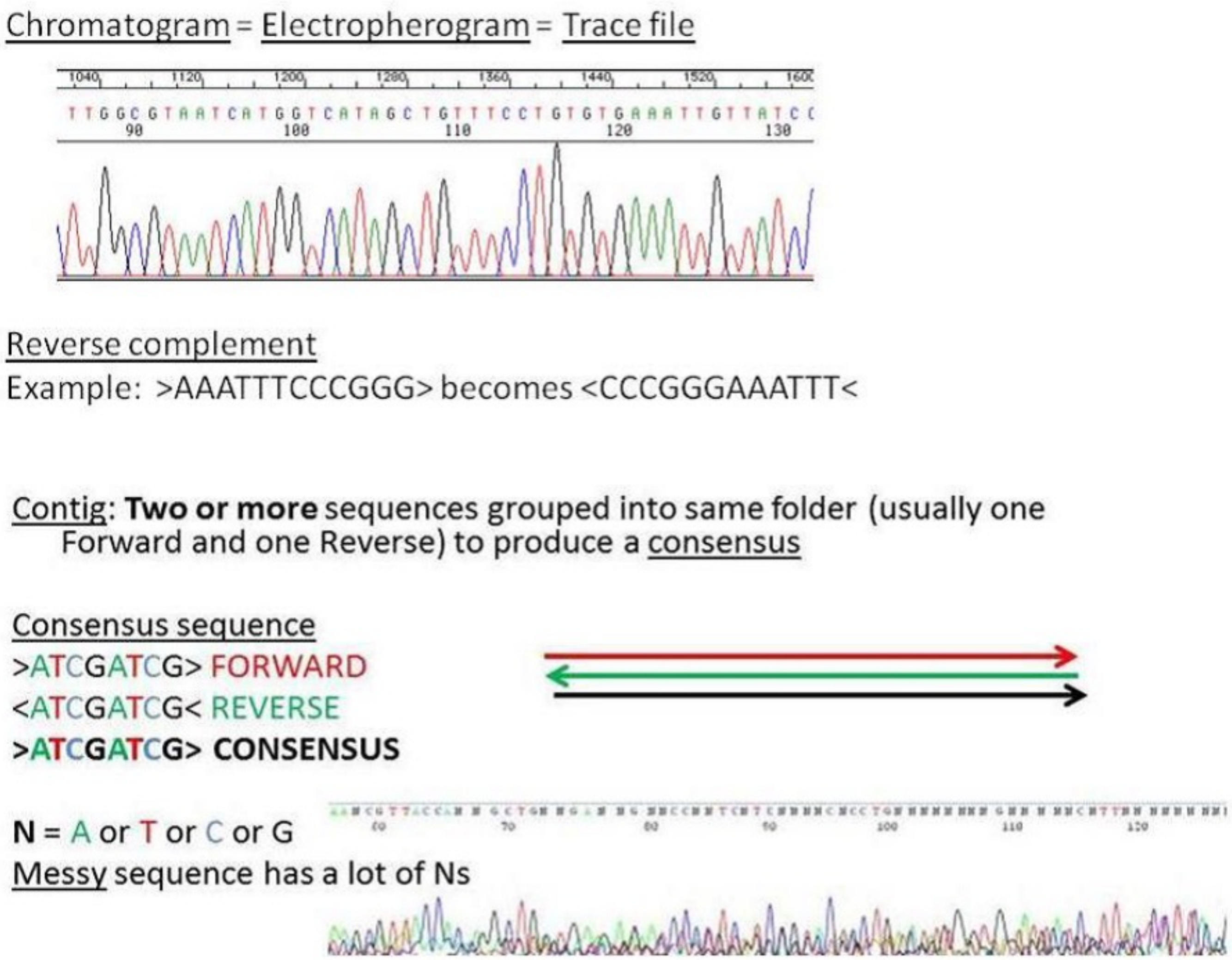
```
writeFasta(my_read,
           outputDir = "output/",
           compress = FALSE)
```

Building Contigs

Forward + Reverse

Sequencing both strands provides confirmation.

A **Contig** is the consensus sequence built by aligning overlapping forward and reverse reads.



Contig Assembly Code

```
# Load reads into a contig my_contig <- SangerContig( parentDirectory = "data/", FASTA_File = "reference.fasta", contigName = "Gene_X" ) # Extract consensus consensusString(my_contig)
```

Generating Reports

Create a comprehensive HTML report for documentation.

```
generateReport(my_read,
               outputDir = "reports/")
```

Includes: Chromatogram, trimming points, quality stats, and sequence text.

Sanger Analysis Best Practices

👀

Visual Check

Always inspect the chromatogram manually.

🛑

Threshold

Use Q20 as the minimum for trusted bases.

✂️

Trim

Always trim poor quality ends.

🔄

Bi-directional

Sequence Fwd & Rev for critical samples.

Key Takeaways

- **Sanger Sequencing** is the gold standard for validation.
- **Phred Scores** (Q) measure accuracy logarithmically.
- **Quality Profile** usually dips at the ends.
- **sangeranalyseR** automates loading, trimming, and reporting.
- **Contigs** combine Fwd/Rev reads for higher confidence.

Tutorial 10

Processing Sanger Data

Open your R environment

Tutorial 10: Processing Sanger Data

Quality Analysis & Trimming

Your Scenario

The Task

You have received Sanger sequencing data for a **viral sample**.

Steps:

1. Simulate data.
2. Analyze quality profile.
3. Calculate stats.
4. Trim low-quality ends.

Step 1: Simulate Data

Run the provided code to generate a data frame with sequence and quality scores.

```
# Data will have:
# Position
# Base (A, T, G, C)
# Quality (Phred Score)
```

Step 2: Visualize Quality

Plot the quality score across the sequence.

```
plot(sanger$Position, sanger$Quality,
     type = "l",
     ylab = "Phred Score")
abline(h = 20, col = "red")
```

Look for the "frown" shape: low ends, high middle.

Step 3: Quality Stats

TODO: Fill in mean_qual <- mean(sanger$...) bases_q20 <- sum(sanger$Quality >= ...)

Step 4: Quality Trimming

Find the first and last position where quality meets the threshold.

```r
# TODO: Find start
trim_start <- which(sanger$Quality >= 20)[1]

# TODO: Find end
trim_end <- max(which(sanger$Quality >= 20))
```

Step 5: Extract Clean Sequence

```
trimmed <- sanger[trim_start:trim_end, ]

# Collapse to string
clean_seq <- paste(trimmed$Base, collapse = "")
```

Compare the original length vs trimmed length. Did you improve the overall quality?

Session 10 Complete

You are ready for Assessment 3.